Generating function for six-cell shaded mesh patterns on separable permutations

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Editorial note. Complete native author body, pattern diagrams, tables and embedded bibliography retained; one original bundled theorem counted per task. No proof repair, added assumptions or mathematical certification.

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Allowed standard prerequisites (editorial).

- Permutation pattern avoidance and generating functions. Native mesh-pattern shading diagrams, separability definitions, symmetry conventions and all supporting decomposition results retained.. The selected coupled pattern-distribution derivation remains in the counted author proof.

1 Introduction

A permutation of length n is a rearrangement of the set $[n] := \{1, 2, \ldots, n\}$. Denote by S_n the set of permutations of $[n]$ and let ε be the empty permutation. A *pattern* is a permutation. A permutation $\pi_1\pi_2\cdots\pi_n \in S_n$ avoids a pattern $p = p_1p_2\cdots p_k \in S_k$ if there is no subsequence $\pi_{i_1}\pi_{i_2}\cdots\pi_{i_k}$ such that $\pi_{i_j} < \pi_{i_m}$ if and only if $p_j < p_m$. Let $S_n(p)$ denote the set of p-avoiding permutations of length n. For a permutation $\pi = \pi_1\pi_2\cdots\pi_n$, its *reverse* is the permutation $\pi^r = \pi_n\pi_{n-1}\cdots\pi_1$ and its *complement* is the permutation $\pi^c = c(\pi_1)c(\pi_2)\cdots c(\pi_n)$ where $c(x) = n + 1 - x$. Also, the length of a permutation π, denoted by $|\pi|$, is the number of elements in π. For example, for $\pi = 423156$, $|\pi| = 6$, $\pi^r = 651324$, and $\pi^c = 354621$. Finally, for a sequence σ of distinct numbers, the *reduced form* of σ, denoted $\mathrm{red}(\sigma)$, is obtained by replacing the i-th smallest number in σ by i. For example, $\mathrm{red}(4295) = 2143$.

The concept of a *mesh pattern* was introduced by Brändén and Claesson [5]. The study of mesh patterns has become a prominent area of research within the theory of patterns in permutations, which has attracted significant attention (see [12]). For enumerative results related to mesh patterns, see, for example, [2, 11, 13, 14, 15, 17, 20].

A mesh pattern of length k is a pair (τ, R), where $\tau \in S_k$ and R is a subset of $[\![0, k]\!] \times [\![0, k]\!]$, where $[\![0, k]\!]$ denotes the interval of integers from 0 to k. Let (i, j) be the box whose corners have coordinates

$(i,j),(i,j+1),(i+1,j+1)$, and $(i+1,j)$, and let the horizontal lines (resp., vertical lines) represent the values (resp., positions) in the pattern. Mesh patterns can be drawn by shading the boxes in R. A mesh pattern with $\tau = 123$ and $R = \{(0,0),(1,2),(2,1),(3,0)\}$ is drawn as follows



Given a permutation $\pi = \pi_1\pi_2\cdots\pi_n \in S_n$, a subsequence $\pi' = \pi_{i_1}\pi_{i_2}\cdots\pi_{i_k}$ is an *occurrence* of a mesh pattern $p = (\tau, R)$ in π if (i) π' is order-isomorphic to τ, and (ii) the shaded regions defined by R contain no elements of π that are not in π'. Otherwise, π is said to avoid p, denoted by $\pi \in S_n(p)$. For example, consider the mesh pattern $p = \big(123, \{(0,0),(1,2),(2,1),(3,1)\}\big)$. The permutation 23154 contains two occurrences of p, whereas the permutation 14325 avoids p. We let $p(\pi)$ denote the number of occurrences of (mesh) pattern p in π.

Two patterns q_1 and q_2 are *Wilf-equivalent*, denoted $q_1 \sim q_2$, if $|S_n(q_1)| = |S_n(q_2)|$. Moreover, q_1 and q_2 are called *equidistributed*, denoted $q_1 \sim_d q_2$, if the number of permutations of length n with k occurrences of q_1 is equal to that with k occurrences of q_2 for any nonnegative integers n and k. The stricter notion of *joint equidistribution* of two patterns, denoted $q_1 \sim_{jd} q_2$, means that the number of n-permutations with k occurrences of q_1 and ℓ occurrences of q_2 is equal to that with ℓ occurrences of q_1 and k occurrences of q_2.

For $n \geq 2$, a permutation $\pi_1\pi_2\cdots\pi_n$ is *irreducible* if there is no i, $2 \leq i \leq n$, such that any element in $\pi_1\cdots\pi_{i-1}$ is less than each element in $\pi_i\cdots\pi_n$. For example, the permutation 42513 is irreducible. By definition, the permutation 1 is irreducible. A permutation is *reducible* if it is not irreducible. The empty permutation ε is neither reducible nor irreducible. Each reducible permutation can be cut into irreducible pieces called *irreducible components*, or just *components*. For example, the components of the permutation 312546978 are 312, 54, 6, and 978.

Suppose $\pi = \pi_1\pi_2\cdots\pi_m \in S_m$ and $\sigma = \sigma_1\sigma_2\cdots\sigma_n \in S_n$. We define the *direct sum* (or simply, *sum*) $\oplus$, and the *skew sum* $\ominus$ by building the permutations $\pi\oplus\sigma$ and $\pi\ominus\sigma$ as follows:

$$(\pi\oplus\sigma)_i = \begin{cases} \pi_i & \text{if } 1 \leq i \leq m, \\ \sigma_{i-m}+m & \text{if } m+1 \leq i \leq m+n, \end{cases}$$

$$(\pi\ominus\sigma)_i = \begin{cases} \pi_i+n & \text{if } 1 \leq i \leq m, \\ \sigma_{i-m} & \text{if } m+1 \leq i \leq m+n. \end{cases}$$

For example, $14325 \oplus 4231 = 143259786$ and $14325 \ominus 4231 = 587694231$.

The *separable permutations* are those which can be built from the permutation 1 by repeatedly applying the $\oplus$ and $\ominus$ operations. Bose, Buss and Lubiw [4] introduced the notion of separable permutation in 1998, and it is well-known [12] that the set of all separable permutations of length $n \geq 1$ is precisely $S_n(2413, 3142)$ (we assume here that by definition, the empty permutation ε is not separable). Separable permutations appear in the literature in various contexts (see [6] for references).

Clearly, the reverse, or complement, or (usual group theoretic) inverse, or any composition of these operations applied to a separable permutation gives a separable permutation. It is not difficult to see that any $\pi \in S_n(2413, 3142)$ has the following structure (see also Figure 1 for a schematic representation, where a permutation is viewed as a diagram with a dot in position (i, π_i) for each element π_i of π):

$$\pi = L_1 L_2 \cdots L_m n R_m R_{m-1} \cdots R_1$$

where

(1) for $1 \le i \le m$, L_i and R_i are non-empty ($\neq \varepsilon$), with possible exception for L_m and R_1, separable permutations which are intervals in π (that is, consist of all elements in $\{a, a+1, \ldots, b\}$ for some a and b);

(2) $R_1 < L_1 < R_2 < L_2 < \cdots < R_m < L_m$, where $A < B$, for two permutations A and B, means that each element of A is less than every element of B. In particular, R_1, if it is not empty, contains 1.

For example, if $\pi = 2165743$ then $L_1 = 21$, $L_2 = 65$, $R_1 = \varepsilon$ and $R_2 = 43$. Note that a separable permutation is irreducible if and only if $R_1 \neq \epsilon$.

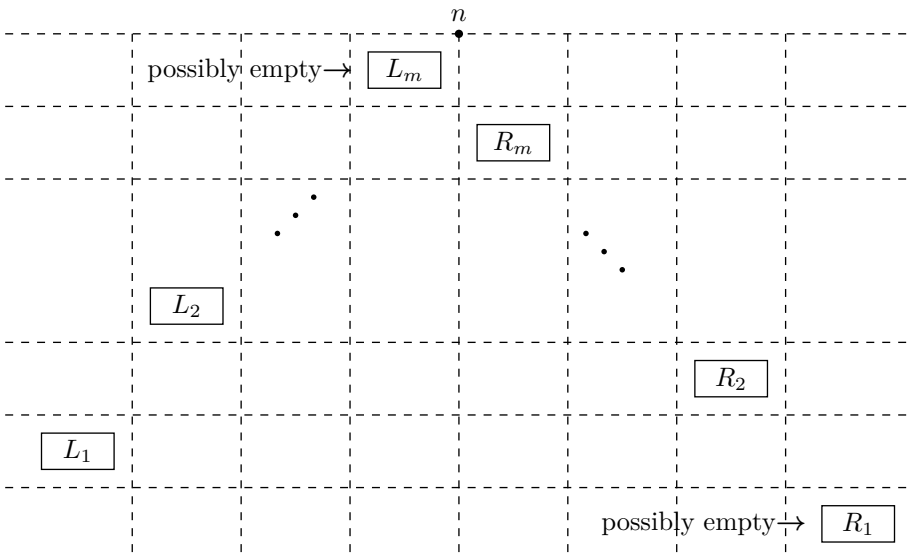


Figure 1: A schematic representation of the permutation diagrams corresponding to separable permutations. Each L_i and R_j is a separable permutation.

Throughout this paper, we use the abbreviation "g.f." to denote a *generating function*. We denote by $F(x)$ the generating function (g.f.) for separable permutations:

$$F(x) := \sum_{n \ge 1} |S_n(3142, 2413)| x^n = \frac{1 - x - \sqrt{1 - 6x + x^2}}{2}. \tag{1}$$

The notation $F(x,t)$, and likewise $F(x, t_1, t_2)$, is local to each theorem and its proof: the marked pattern or patterns are specified there. Throughout the paper, $F(x)$ always denotes the series in (1).

1.1 Symmetry operations

Clearly, applying the reverse, complement, or (usual group-theoretic) inverse operation, or any composition of these operations, to a separable permutation yields a separable permutation. For a given mesh pattern (τ, R) of length n, we define

$$(\tau, R)^r = (\tau^r, R^r), \quad (\tau, R)^c = (\tau^c, R^c), \quad (\tau, R)^i = (\tau^i, R^i),$$

where τ^i is the usual inverse, and $R^r = \{(n-x, y) : (x, y) \in R\}$, $R^c = \{(x, n-y) : (x, y) \in R\}$, and $R^i = \{(y, x) : (x, y) \in R\}$.

Hence, *reverse* is a reflection around the vertical center line, *complement* is a reflection around the horizontal center line, and *inverse* is the reflection around the southwest-to-northeast diagonal. Figure 2 illustrates applications of these operations on the pattern $p = (312, \{(0,1), (1,3), (2,2)\})$.

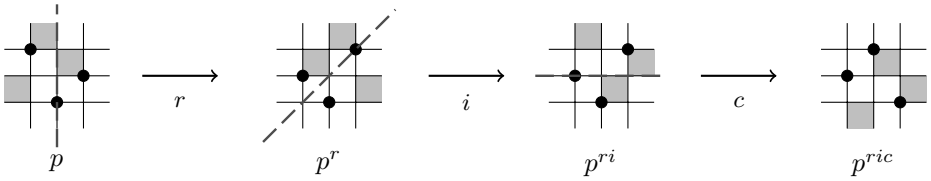


Figure 2: Several symmetries of a mesh pattern.

1.2 Distributions of statistics on separable permutations

For a permutation $\pi = \pi_1\pi_2\cdots\pi_n$, the *descent* (resp., *ascent*) statistic on π, $\mathrm{des}(\pi)$ (resp., $\mathrm{asc}(\pi)$), is defined as the number of $i \in \{1, 2, \ldots, n-1\}$ such that $\pi_i > \pi_{i+1}$ (resp., $\pi_i < \pi_{i+1}$). For example, $\mathrm{des}(561423) = 2$ and $\mathrm{asc}(25134) = 3$.

The following lemma will be used in Theorem 17 below. It gives the distribution of descents (equivalently, by using reverse, ascents) on separable permutations.

Lemma 1 ([8, Cor. 2.4]). *The g.f.*

$$Q(x,t) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\mathrm{asc}(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\mathrm{des}(\pi)}$$

satisfies the functional equation

$$tQ^3(x,t) + txQ^2(x,t) + ((1+t)x - 1)Q(x,t) + x = 0. \tag{2}$$

For a permutation $\pi = \pi_1\pi_2\cdots\pi_n$, π_i is a *right-to-left maximum* (resp., *right-to-left minimum*) if $\pi_i > \pi_j$ (resp., $\pi_i < \pi_j$) for all $j > i$. In particular, π_n is a right-to-left maximum and a right-to-left minimum. The number of right-to-left maxima (resp., right-to-left minima) is denoted by $\mathrm{rmax}(\pi)$ (resp., $\mathrm{rmin}(\pi)$). For example, $\mathrm{rmax}(436512) = 3$ and $\mathrm{rmin}(326154) = 2$. Also, π_i is a *left-to-right maximum* (resp., *left-to-right minimum*), denoted by $\mathrm{lmax}(\pi)$ (resp., $\mathrm{lmin}(\pi)$) if $\pi_i > \pi_j$ (resp., $\pi_i < \pi_j$) for all $j < i$. In particular, π_1 is a left-to-right maximum and a left-to-right minimum. For example, $\mathrm{lmax}(325164) = \mathrm{lmin}(326154) = 3$.

In this paper, we will frequently make use of the following theorem from [6], which describes the distributions of four permutation statistics on separable permutations, as well as on their reducible and irreducible subclasses.

Theorem 1 ([6, Thm. 5]). *The g.f.* $G(x,t) := \sum_\pi x^{|\pi|} t^{\mathrm{stat}(\pi)}$ *giving the distribution of any statistic* $\mathrm{stat} \in \{\mathrm{lmax}, \mathrm{rmax}, \mathrm{lmin}, \mathrm{rmin}\}$ *on separable permutations is*

$$\frac{4\sqrt{\left(\frac{x}{4} + \frac{5}{4} - \frac{1}{4}\sqrt{x^2-6x+1} - tx\right)^2 + \sqrt{x^2-6x+1} - x - 1} - \sqrt{x^2-6x+1} + 4tx + x - 3}{2\left(\sqrt{x^2-6x+1} - x - 1\right)}. \tag{3}$$

The distributions of stat *on irreducible and reducible separable permutations are given in Table 1.*

g.f.	irreducible	reducible
$\frac{G^2(x,t)}{1+G(x,t)} + xt$	stat $\in \{\mathrm{rmax}, \mathrm{lmin}\}$	stat $\in \{\mathrm{lmax}, \mathrm{rmin}\}$
$\frac{G(x,t)}{1+G(x,t)} - xt$	stat $\in \{\mathrm{lmax}, \mathrm{rmin}\}$	stat $\in \{\mathrm{rmax}, \mathrm{lmin}\}$

Table 1: Distributions of statistics in Theorem 1.

1.3 Distribution of mesh patterns in the literature

Kitaev and Zhang [14] initiated a systematic study of the distributions of short mesh patterns (of length at most 2) in permutations, extending the earlier work of [11] on the systematic study of mesh pattern avoidance in permutations. These investigations were later extended to certain infinite families of mesh patterns in [15]. The systematic study of mesh patterns of length 3 was initiated in [17] and subsequently continued in [18, 19]. Other contributions to the literature on distributions of mesh patterns include [7, 9, 10]. The distribution of mesh patterns on a particular subclass of permutations, namely *king permutations*, was studied in [16].

1.4 Results in this paper

In this paper, we first determine the distributions of all mesh patterns of length 1 (see Table 2 and the references therein). In one instance, Theorem 7, we derive the joint equidistribution of strict fixed points and reverse strict fixed points. We then classify the mesh patterns 12 and 21 with identical shadings into 31 equivalence classes based on their distributions; these classes are listed in Tables 3–5, with further mergers described in Table 7. These results are obtained using symmetry operations as well as by explicitly determining distributions of mesh patterns (we derived 24 distributions). In many cases, we are also able to determine the joint distributions of the corresponding pairs. Interestingly, although the distributions of the patterns  and  are the same, the joint distributions of the pairs {  ,  } and {  ,  } are different.

2 Distribution of mesh pattern of length 1 on separable permutations

There are $2^4 = 16$ mesh patterns of length 1. Under inverse, reverse, and complement, these patterns fall into six equidistribution classes, listed in Table 2. For each class, we determine the corresponding distribution. In addition, we compute the joint distribution of the patterns  and  . In Table 2, mesh patterns that are equidistributed by the reverse, complement, or inverse operations, are grouped together within the same frame.

In a separable permutation, occurrences of the patterns  ,  ,  , and  correspond, respectively, to left-to-right maxima, left-to-right minima, right-to-left maxima, and right-to-left minima. Consequently, we can use Theorem 1 to obtain the following result, which establishes the distributions of the mesh patterns N1–N4.

Nr.	Pattern	Ref.	Nr.	Pattern	Ref.	Nr.	Pattern	Ref.
N1		Thm. 2	N5		Thm. 5	N9		Thm. 6
N2			N6			N10		
N3			N7			N11		
N4			N8			N12		
N13		Thm. 7	N15		Thm. 3	N16		Thm. 4
N14								

Table 2: Symmetry-induced equidistribution classes for mesh patterns of length 1.

Theorem 2. *For $p \in \{$  ,  ,  ,  $\}$, and $G(x,t)$ given by (3), we have*

$$\sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{p(\pi)} = G(x,t).$$

The proof of the following theorem is immediate from the fact that each element in a permutation contributes an occurrence of the pattern N15.

Theorem 3. *We have*

$$\sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{}(\pi)} = F(xt) = \frac{1-tx-\sqrt{1-6tx+t^2x^2}}{2}.$$

Theorem 4 (Pattern N16)**.** *We have*

$$F(x,t) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \frac{1+(2t-3)\,x-\sqrt{1-6x+x^2}}{2}.$$

Proof. Clearly, the only separable permutation that contains the pattern  is 1, so

$$F(x,t) = tx + F(x) - x. \tag{4}$$

By substituting (1) into (4), we obtain the desired result. □

Theorem 5 (Patterns N5–N8)**.** *For $p \in \{$  ,  ,  ,  $\}$, we have*

$$F(x,t) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{p(\pi)} = \frac{t(1-x-\sqrt{1-6x+x^2})}{2}.$$

Proof. W.l.o.g., consider N5=  . It occurs exactly once, as the leftmost element, in any separable permutation, so $F(x,t) = tF(x)$, and the result follows by (1). □

Theorem 6 (Patterns N9–N12). *For* $p \in \{$  ,  ,  ,  $\}$*, we have*

$$F(x,t) := \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{p(\pi)}$$
$$= \frac{1}{2}\Big(1 - x - \sqrt{1-6x+x^2} - (t-1)\,x\left(x - 3 + \sqrt{1-6x+x^2}\right)\Big).$$

Proof. W.l.o.g., consider the pattern N9 =  . Only the element 1 occurring at the end of a permutation can be an occurrence of the pattern  , and in this position, 1 cannot be part of an occurrence of the patterns 2413 or 3142. Hence, if $A(x)$ denotes the g.f. for separable permutations that end with their smallest element, we have

$$F(x,t) = tA(x) + F(x) - A(x). \tag{5}$$

Moreover, any permutation counted by $A(x)$ can be generated by taking any separable permutation or the empty permutation and adjoining the new smallest element to its right, so $A(x) = x\,(F(x)+1)$. Together with (1) and (5), this gives the desired result. □

Occurrences of the pattern  are known as *strict fixed points*. The avoidance of this pattern on permutations was studied in [3], and its distribution in [14]. We extend the study of the distribution of strict fixed points to separable permutations. Since separable permutations are invariant under the reverse operation, the distribution of the pattern N13 =  coincides with that of the pattern N14 =  . Moreover, we obtain the joint distribution of the patterns  and  . The following theorem establishes the distributions of the mesh patterns N13 and N14, as well as their joint distribution.

Theorem 7 (Patterns N13 and N14). *We have*  $\sim_{jd}$  . *Moreover, if*

$$F(x,t) := \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{\#\text{}(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{\#\text{}(\pi)} \quad \textit{and}$$

$$F(x,t_1,t_2) := \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t_1^{\#\text{  }(\pi)} t_2^{\#\text{  }(\pi)},$$

then

$$F(x,t) = \frac{1 + (4t-3)x - \sqrt{1+(x-6)x}}{3 + (3-4t)x + \sqrt{1+(x-6)x}}; \tag{6}$$

$$F(x,t_1,t_2) = \frac{1}{4}\Bigg(4t_1t_2x + \frac{\left(-1 + (3-4t_1)x + \sqrt{1-6x+x^2}\right)^2}{3 + (3-4t_1)x + \sqrt{1-6x+x^2}} + \frac{\left(-1 + (3-4t_2)x + \sqrt{1-6x+x^2}\right)^2}{3 + (3-4t_2)x + \sqrt{1-6x+x^2}}\Bigg). \tag{7}$$

Proof. The joint equidistribution follows by applying the reverse operation that swaps occurrences of the patterns. Then we utilize the structure of a separable permutation π of length n shown in Figure 1 and consider two cases, depending on whether π is reducible or not.

Case 1. In the reducible case (see Figure 3a), the block B must be nonempty, and can be any separable permutation. Case 1 can be divided into two subcases:

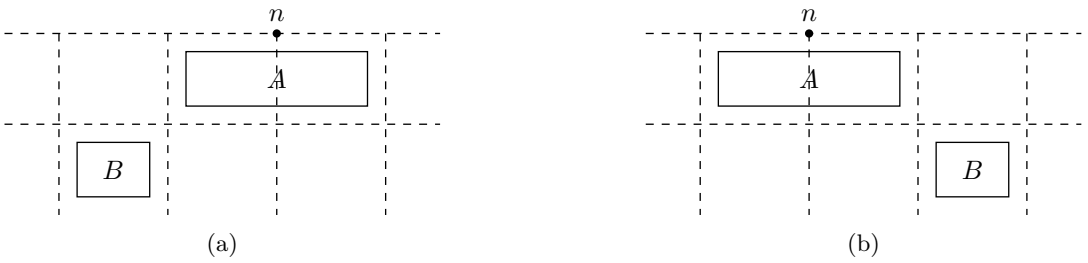


Figure 3: Related to the proof of Theorem 7.

(a) $A = \varepsilon$ and $B \neq \varepsilon$. In this subcase, we obtain the term $xt_1F(x, t_1)$, since the element n contributes an occurrence of the pattern  but not of  . Moreover, because of n, no element in B can be an occurrence of  , while n cannot destroy any occurrence of the pattern  in B.

(b) $A \neq \varepsilon$ and $B \neq \varepsilon$. In this subcase, to be disjoint from (a), A has at least one element to the right of n, and by the structure shown in Figure 1, the minimum element m of A lies to the right of n. Therefore, in this subcase we obtain the term

$$\frac{F(x) - x}{2} F(x, t_1).$$

Indeed,

- B contributes the term $F(x, t_1)$, since, because of n, it cannot contribute any occurrences of the pattern  , and is otherwise independent of the rest of π.
- Since $B \neq \varepsilon$, the rest of π cannot contribute any occurrences of  . Moreover, because of m (resp., n), no element to the left (resp., to the right) of n, including n itself, can contribute an occurrence of the pattern  . Therefore, A together with n, being an irreducible permutation of length at least 2, contributes the term $\frac{F(x)-x}{2}$ (by reversal symmetry, half of separable permutations are reducible and half are irreducible).

Case 2. In the irreducible case (see Figure 3b), the block B is nonempty, except in the trivial case $n = 1$, and can be any separable permutation. We consider three subcases.

(a) $A = B = \varepsilon$, then $\pi = 1$ giving the term of t_1t_2x.

(b) $A = \varepsilon$ and $B \neq \varepsilon$, which gives $xt_2\, F(x, t_2)$ by reversing the permutations in Case 1(a).

(c) $A \neq \varepsilon$ and $B \neq \varepsilon$, which gives $\frac{F(x)-x}{2} F(x, t_2)$ by reversing the permutations in Case 1(b).

Combining the two cases above, we obtain

$$F(x, t_1, t_2) = xt_1t_2 + xt_1\, F(x, t_1) + xt_2\, F(x, t_2) + \frac{F(x) - x}{2} \big(F(x, t_1) + F(x, t_2)\big). \tag{8}$$

Letting $t_2 = 1$ in (8), we get

$$F(x, t_1) = xt_1 + xt_1\, F(x, t_1) + x\, F(x) + \frac{F(x) - x}{2} \big(F(x, t_1) + F(x)\big). \tag{9}$$

Substituting (1) into (9), solving for $F(x, t_1)$, and then replacing t_1 by t yields (6). Finally, substituting $F(x, t_1)$ and $F(x, t_2)$, both given by (6), into (8) yields (7). □

3 Equidistribution of mesh patterns of length 2 with the same shading on separable permutations

There are $n! \cdot 2^{(n+1)^2}$ mesh patterns of length n, and for $n = 2$ there are already 1024 distinct patterns, among which $2^9 = 512$ have distinct shadings. Computer experiments indicate that, on separable permutations, when the patterns 12 and 21 are regarded as mesh patterns with selectively shaded boxes, at most 124 equidistributed pairs can exist.

In this section, we employ symmetry operations and generating function methods to prove that the 124 pairs of mesh patterns presented in Tables 3–6 are equidistributed. These pairs are further classified into 38 distinct symmetry-induced classes according to the corresponding symmetry operations.

3.1 Symmetry operations

Out of the 124 equidistributed pairs obtained, 112 pairs can be naturally derived through (composition of) symmetry operations. The explicit correspondences and symmetry-induced groupings of these pairs are presented in Tables 3–5.

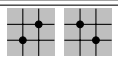
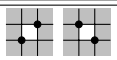
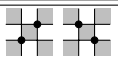
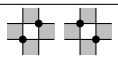
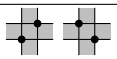
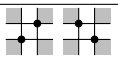
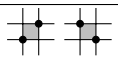
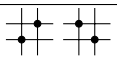
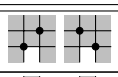
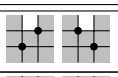
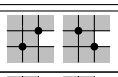
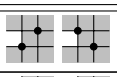
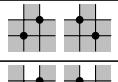
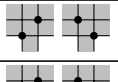
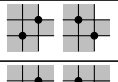
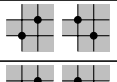
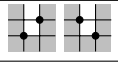
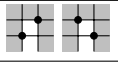
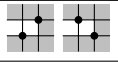
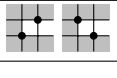
Nr.	Patterns	Ref.	Nr.	Patterns	Ref.	Nr.	Patterns	Ref.
S1		Thm. 10	S2		Thm. 11	S3		
S4			S5		Thm. 21	S6		
S7			S8					

Table 3: Symmetry-induced equidistribution classes of mesh patterns with a single pattern per class, and references to their known enumeration.

For the patterns 12 and 21, both the reverse and the complement operations transform one into the other; that is, applying either operation maps 12 to 21, and vice versa. In all pairs of mesh patterns listed in Tables 3–5, the shaded regions are symmetric with respect to the horizontal or vertical axis, corresponding to the complement and reverse operations, respectively. Consequently, for each of the 112 pairs of mesh patterns shown in these three tables, the two patterns within each pair are equidistributed under the action of the reverse or complement symmetry.

In Table 3, the single pair of mesh patterns is invariant under all three symmetries. In Table 4, each frame contains two such pairs that can be mapped to each other by one of the inverse, complement, or reverse operations. Similarly, in Table 5, each frame comprises four pairs of mesh patterns that can be mutually mapped and readily verified via these three symmetries or their compositions.

Nr.	Patterns	Nr.	Patterns	Nr.	Patterns	Nr.	Patterns	Ref.
S17		S18		S19		S20		Thm. 14
S21		S22		S23		S24		Thm. 15
S25		S26		S27		S28		Thm. 15

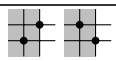
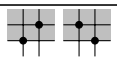
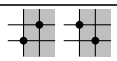
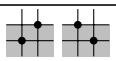
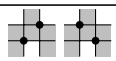
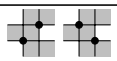
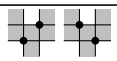
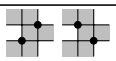
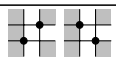
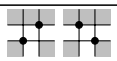
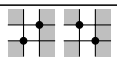
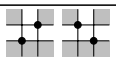
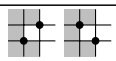
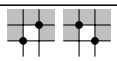
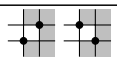
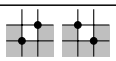
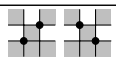
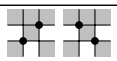
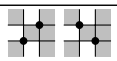
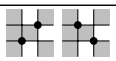
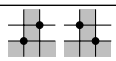
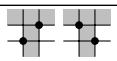
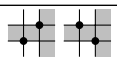
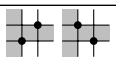
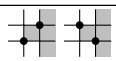
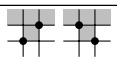
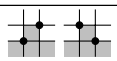
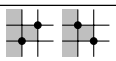
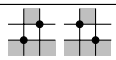
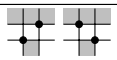
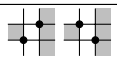
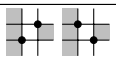
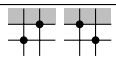
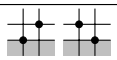
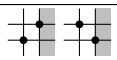
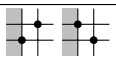
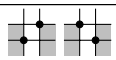
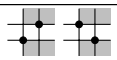
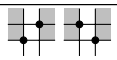
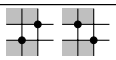
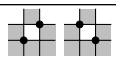
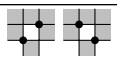
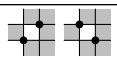
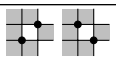
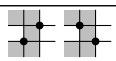
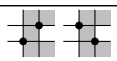
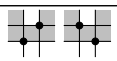
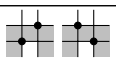
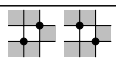
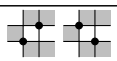
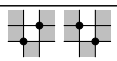
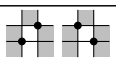
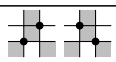
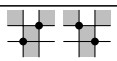
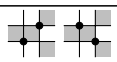
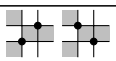
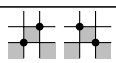
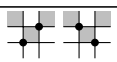
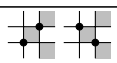
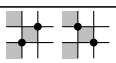
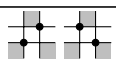
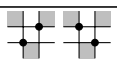
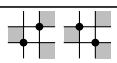
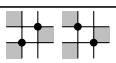
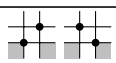
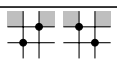
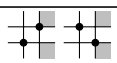
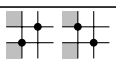
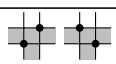
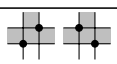
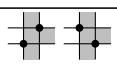
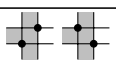
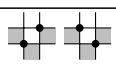
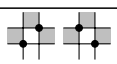
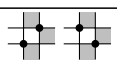
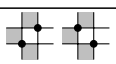
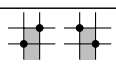
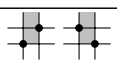
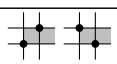
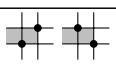
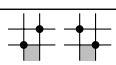
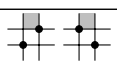
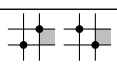
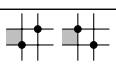
Nr.	Patterns	Nr.	Patterns	Nr.	Patterns	Nr.	Patterns	Ref.
S29		S30		S31		S32		Thm. 12
S33		S34		S35		S36		Thm. 22
S37		S38		S39		S40		Thm. 23
S41		S42		S43		S44		Thm. 24
S45		S46		S47		S48		Thm. 18
S49		S50		S51		S52		Thm. 16
S53		S54		S55		S56		Thm. 25
S57		S58		S59		S60		Thm. 26
S61		S62		S63		S64		Thm. 19
S65		S66		S67		S68		Thm. 20
S69		S70		S71		S72		Thm. 27
S73		S74		S75		S76		Thm. 28
S77		S78		S79		S80		Thm. 29
S81		S82		S83		S84		
S85		S86		S87		S88		
S89		S90		S91		S92		
S93		S94		S95		S96		
S97		S98		S99		S100		
S101		S102		S103		S104		
S105		S106		S107		S108		
S109		S110		S111		S112		

Table 5: Symmetry-induced equidistribution classes of mesh patterns, with four patterns per class, and references to their known enumeration.

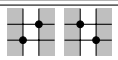
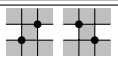
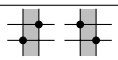
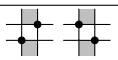
Nr.	Patterns	Nr.	Patterns	Ref.	Nr.	Patterns	Nr.	Patterns	Ref.
S9		S10		Thm. 12	S11		S12		Thm. 13
S13		S14		Thm. 17	S15		S16		

Table 4: Symmetry-induced equidistribution classes of mesh patterns with two patterns per class, and references to their known enumeration.

3.2 The generating functions method

Unlike the other pairs of mesh patterns, the equidistribution of each pair in Table 6 cannot be established solely by symmetry arguments. Instead, we employ generating function techniques to provide a proof.

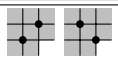
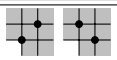
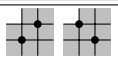
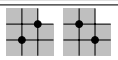
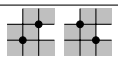
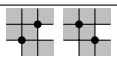
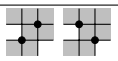
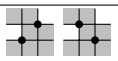
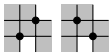
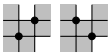
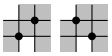
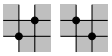
Nr.	patterns	Nr.	Patterns	Nr.	Patterns	Nr.	Patterns	Ref.
S113		S114		S115		S116		Thm. 8
S117		S118		S119		S120		Thm. 9
S121		S122		S123		S124		

Table 6: Symmetry-induced equidistribution classes of mesh patterns proved via generating functions.

In Table 6, pairs of mesh patterns sharing the same equidistribution are grouped within the same frame. It suffices to prove equidistribution for only one pair of patterns in each frame; the remaining pairs are obtained by the inverse, reverse, and complement operations.

Theorem 8 (Pairs S113–S116). *We have*  $\sim_{jd}$  . *Moreover,*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} \\ &= \frac{1}{2}\Big(1-x-\sqrt{1-6x+x^2}+(t-1)x^2\big(3-x-\sqrt{1-6x+x^2}\big)\Big); \end{aligned} \tag{10}$$

$$\begin{aligned} F(x,t_1,t_2) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\text{  }(\pi)} t_2^{\text{  }(\pi)} \\ &= \frac{1}{2}\Big(1-x-\sqrt{1-6x+x^2}+(t_1+t_2-2)x^2\big(3-x-\sqrt{1-6x+x^2}\big)\Big). \end{aligned}$$

Proof. Clearly, if a separable permutation $\pi = \pi_1\cdots\pi_n$ has an occurrence of the pattern  (resp., ), then $\pi_1\pi_2 = (n-1)n$ (resp., $\pi_1\pi_2 = n(n-1)$) is the only occurrence of the pattern in π, and $\pi_3\cdots\pi_n$ is any separable permutation in S_{n-2}. Therefore, the map that swaps n and $n-1$ in all permutations of length n establishes the joint equidistribution  $\sim_{jd}$  .

Consider the pattern  , and let $A(x,t)$ be the g.f. for permutations with one occurrence of the pattern. Then, from the above, $A(x,t) = tx^2(F(x)+1)$, and

$$F(x,t) = A(x,t) + (F(x) - A(x,1)) = (t-1)x^2(F(x)+1) + F(x). \tag{11}$$

Substituting (1) into (11) yields (10).

Finally, because both patterns cannot occur in the same permutation, we have

$$\begin{aligned} F(x,t_1,t_2) &= A(x,t_1) + A(x,t_2) + (F(x) - 2A(x,1)) \\ &= (t_1+t_2-2)x^2(F(x)+1) + F(x), \end{aligned} \tag{12}$$

and the formula for $F(x,t_1,t_2)$ is obtained by substituting (1) into (12). □

Theorem 9 (Pairs S117–S124)**.** *We have*  $\sim_{jd}$  *. Moreover,*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} \\ &= \frac{1}{4}\Big(tx^2(x-3+\sqrt{1-6x+x^2})^2 \\ &\qquad -x^2(x-3+\sqrt{1-6x+x^2})^2 - 2(x-1+\sqrt{1-6x+x^2})\Big); \end{aligned} \tag{13}$$

$$\begin{aligned} F(x,t_1,t_2) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\text{  }(\pi)} t_2^{\text{  }(\pi)} \\ &= F(x) + x^2\Big((t_1+t_2)(1-x) + (t_1t_2+1)x - 2\Big)(F(x)+1)^2, \end{aligned} \tag{14}$$

where $F(x)$ is given by (1).

Proof. Clearly, if a separable permutation π contains an occurrence of the pattern  (resp., ), then for some element m, we can write $\pi = A1mB$ (resp., $\pi = Am1B$), where each element of A, if any, is larger than m and each element of B, if any, is smaller than m. Moreover, each mesh pattern can occur at most once in π, and the reduced forms of A and B can be any separable permutations (π is separable if and only if both A and B are separable). Therefore, if $A(x,t)$ is the g.f. for permutations with exactly one occurrence of either pattern, then

$$F(x,t) = A(x,t) + (F(x) - A(x,1)),$$

and

$$A(x,t) = tx^2(F(x)+1)^2, \tag{15}$$

from which, together with (1), we obtain (13) and  $\sim_d$  .

To deal with the joint distribution, first observe that the only way both patterns can occur in the same permutation π is if $\pi = A\,(m+1)\,1\,m\,B$, where every element in A, if any, is larger than $m+1$, and every element in B, if any, is smaller than m. The g.f. for such permutations is clearly $t_1t_2x^3(F(x)+1)^2$. Now, the g.f. for permutations containing  (resp., ), but not  (resp., ), is $A(x,t_1) -$

$t_1x^3(F(x)+1)^2$ (resp., $A(x,t_2)-t_2x^3(F(x)+1)^2$). Moreover, the g.f. for separable permutations avoiding both patterns is

$$F(x) - (2A(x,1) - x^3(F(x)+1)^2).$$

Combining the four generating functions we have that $F(x,t_1,t_2)$ is given by

$$A(x,t_1) + A(x,t_2) - (t_1+t_2)x^3(F(x)+1)^2 + t_1t_2x^3(F(x)+1)^2 + F(x) - (2A(x,1) - x^3(F(x)+1)^2),$$

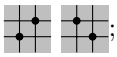
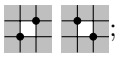
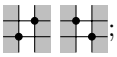
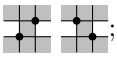
and formula (14) is obtained by applying (15). Since $F(x,t_1,t_2) = F(x,t_2,t_1)$, we have  $\sim_{jd}$  . □

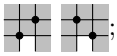
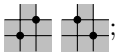
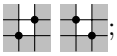
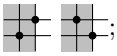
4 Distribution of mesh pattern of length 2 on separable permutations

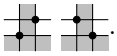
In this section, we determine the distributions of 21 increasing (resp., decreasing) mesh patterns of length 2 on separable permutations. For the pairs of mesh patterns presented in Tables 3–5, we select the first pair of patterns in each class (that is, within each frame) as the representative pair for determining the corresponding distribution, since the equidistribution of all pairs has already been established. In our analysis, we always choose the pattern 12 as the representative.

4.1 "Trivial" distributions

By a "trivial" distribution we mean the situation when either the pattern in question can occur at most once, or the pattern's occurrences can be easily inferred from the shape of the pattern. There are nine classes of pairs of such patterns:

S1 =  ; S2 =  ; S9 =  ; S11 =  ;

S17 =  ; S21 =  ; S25 =  ; S29 =  ;

S49 =  .

Theorem 10 (Pair S1)**.** *We have*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} \\ &= \frac{1}{2}\left(1-x+2(t-1)x^2 - \sqrt{1-6x+x^2}\right). \end{aligned} \tag{16}$$

$$F(x,t_1,t_2) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\text{  }(\pi)} t_2^{\text{  }(\pi)} = (t_1+t_2-2)x^2 + \frac{1-x-\sqrt{1-6x+x^2}}{2}.$$

Proof. The only separable permutation that contains the pattern  is 12. Hence,

$$F(x,t) = tx^2 + F(x) - x^2. \tag{17}$$

By substituting (1) into (17), we obtain (16). Also, it is clear that

$$F(x,t_1,t_2) = t_1x^2 + t_2x^2 + F(x) - 2x^2,$$

which, together with (1), leads to the desired result. □

Theorem 11 (Pair S2). *Theorem 8 also holds for the mesh patterns*  *and*  .

Proof. The pattern  can occur at most once in any permutation $\pi = \pi_1 \cdots \pi_n$. In this case, we must have $\pi_1 = 1$ and $\pi_n = n$ and $\mathrm{red}(\pi_2 \cdots \pi_{n-1})$ can be any separable permutation of length $n-2$. If $A(x,t)$ is the g.f. of separable permutations with one occurrence of the pattern, then all formulas and functional relations in the proof of Theorem 8 are exactly the same, leading to the desired result. □

The following lemma is an immediate corollary of the fact that separable permutations are closed under reversal and under the complement.

Lemma 2. *For every $n \geq 2$, exactly half of the separable permutations of length n have their first element larger than their last element. Likewise, exactly half begin with an ascent, and half begin with a descent.*

Theorem 12 (Pairs S9, S10, S29–S32). *We have*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{}(\pi)} \\ &= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{}(\pi)} \\ &= x + \frac{1}{4}(1+t)\left(1-3x-\sqrt{1-6x+x^2}\right). \end{aligned} \tag{18}$$

$$\begin{aligned} F(x,t_1,t_2) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\text{  }(\pi)} t_2^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\text{  }(\pi)} t_2^{\text{  }(\pi)} \\ &= x + \frac{1}{4}(t_1+t_2)\left(1-3x-\sqrt{1-6x+x^2}\right). \end{aligned}$$

Proof. The pattern  can occur at most once in a permutation, in which case the first element is smaller than the last one. If $A(x,t)$ denotes the generating function for separable permutations of length at least 2 with exactly one occurrence of the pattern, then, by Lemma 2, we have

$$A(x,t) = t\,\frac{F(x)-x}{2}. \tag{19}$$

Substituting (19) into (11) (which also holds for the pattern ), we obtain

$$F(x,t) = \frac{(t+1)(F(x)-x)}{2} + x. \tag{20}$$

Furthermore, substituting (1) into (20) yields (18). The formula for the joint distribution follows from the easy to see equation $F(x,t_1,t_2) = x + A(x,t_1) + A(x,t_2)$.

Finally, the arguments for the pattern  are essentially identical, with the only difference being that an occurrence of this pattern must be formed by the leftmost two elements, which form an ascent. □

Interestingly, although the distribution of the mesh pattern  in the next theorem is the same as that of the mesh pattern  , the joint distributions of these patterns within their respective pairs are different. This is because, in one case, both patterns in the pair can occur in the same permutation, whereas in the other case this is not possible.

Theorem 13 (Pairs S11 and S12)**.** *We have*  $\sim_{jd}$  *. Moreover,*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} \\ &= \frac{1}{4}\Big(tx^2(x-3+\sqrt{1-6x+x^2})^2 \\ &\quad -x^2(x-3+\sqrt{1-6x+x^2})^2 - 2(x-1+\sqrt{1-6x+x^2})\Big); \end{aligned} \tag{21}$$

$$\begin{aligned} F(x,t_1,t_2) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\text{  }(\pi)} t_2^{\text{  }(\pi)} \\ &= (t_1+t_2-2)x^2(F(x)+1)^2 + F(x), \end{aligned} \tag{22}$$

where $F(x)$ is given by (1).

Proof. An occurrence of the pattern  in a permutation $\pi = \pi_1\cdots\pi_n$ can only be formed by the smallest and largest elements. In this case, $\pi = A1nB$, where every element in A, if any, is smaller than every element in B, if any, in order to avoid the pattern 3142. There cannot be more than one occurrence of the pattern. If $A(x,t)$ is the g.f. for the number of separable permutations with one occurrence of the pattern, then clearly $A(x,t)$ satisfies (15) (there are $F(x)+1$ ways to generate A, and independently the same number of ways to generate B). Then, (21) can be obtained using the same arguments as those in the proof of Theorem 9.

To complete the proof, we observe that

$$F(x,t_1,t_2) = A(x,t_1) + A(x,t_2) + (F(x) - 2A(x,1)) = (t_1+t_2-2)x^2(F(x)+1)^2 + F(x).$$

□

Theorem 14 (Pairs S17–S20)**.** *The formulas in Theorem 8 also hold for pairs S17–S20.*

Proof. It is sufficient to consider the pair S17. The pattern  can occur at most once in any permutation $\pi = \pi_1\cdots\pi_n$. In this case, we must have $\pi_1 = n-1$ and $\pi_n = n$ and $\mathrm{red}(\pi_2\cdots\pi_{n-1})$ can be any separable permutation of length $n-2$. If $A(x,t)$ is the g.f. of separable permutations with one occurrence of the pattern, then all formulas and functional relations in the proof of Theorem 8 are exactly the same, giving the desired result. □

Theorem 15 (Pairs S21–S28)**.** *We have*

$$F(x,t) = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)}$$

$$= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\boxed{}(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\boxed{}(\pi)}$$
$$= \frac{1}{2}\left(1+(t-1)x\right)\left(1-x-\sqrt{1-6x+x^2}\right). \tag{23}$$

$$F(x,t_1,t_2) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\boxed{}(\pi)} t_2^{\boxed{}(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\boxed{}(\pi)} t_2^{\boxed{}(\pi)}$$
$$= \frac{1}{2}\left(1+(t_1+t_2-2)x\right)\left(1-x-\sqrt{1-6x+x^2}\right).$$

Proof. If the mesh pattern $\boxed{}$ occurs in a permutation π, then the occurrence must be unique and formed by the elements 1 and 2 placed consequently in increasing order. Let π' be the permutation obtained from π by removing the element 2 and decreasing each element larger than 1 by 1. Clearly, π is separable if and only if π' is separable. Hence, if the g.f. for the number of separable permutations with one occurrence of the pattern is $A(x,t)$, then

$$A(x,t) = txF(x), \tag{24}$$

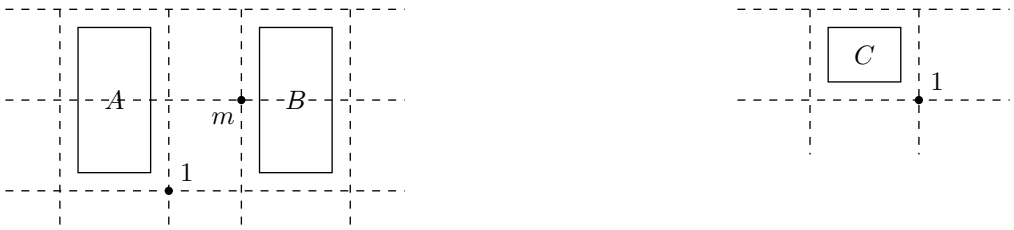
where $F(x)$ counts the choices for π' and x corresponds to the element 2. Substituting (24) into (11) (which also holds for the pattern $\boxed{}$), we obtain

$$F(x,t) = (t-1)xF(x) + F(x). \tag{25}$$

Furthermore, substituting (1) into (25) yields (23). Finally, the formula for the joint distribution follows from

$$F(x,t_1,t_2) = A(x,t_1) + A(x,t_2) + (F(x) - 2A(x,1)).$$

On the other hand, if the mesh pattern $\boxed{}$ occurs in a permutation $\pi = \pi_1\cdots\pi_n$, $n \geq 2$, then the occurrence must be unique and formed by the elements $\pi_1 = 1$ and π_n. Let π'' be the permutation obtained from π by removing the element 1 and decreasing every other element by 1. Clearly, π is separable if and only if π'' is separable. Hence, the g.f. for the number of separable permutations with one occurrence of the pattern is exactly $A(x,t)$ satisfying (24), and all other formulas derived by us for the pattern $\boxed{}$ remain exactly the same, completing our proof. □



(a) (b)

Figure 4: Related to the proof of Theorem 16.

Theorem 16 (Pairs S49–S52)**.** *We have*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} \\ &= \frac{1}{2}\Big(t\left(1+x^2-\sqrt{1-6x+x^2}-x(4-\sqrt{1-6x+x^2})\right) \\ &\quad +x\left(3-x-\sqrt{1-6x+x^2}\right)\Big). \end{aligned} \tag{26}$$

$$\begin{aligned} F(x,t_1,t_2) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t_1^{\text{  }(\pi)} t_2^{\text{  }(\pi)} \\ &= (1-t_1t_2)x+\frac{1}{2}\Big((t_1+t_2)x+t_1t_2(1-2x)\Big)(1-x-\sqrt{1-6x+x^2}). \end{aligned}$$

Proof. By inspecting the pattern  , we observe that the smallest element 1, together with its adjacent element to the right (if it exists), forms a pattern  , as illustrated in Figure 4a. Therefore, a permutation avoids this pattern if and only if it ends with the smallest element 1, as shown in Figure 4b, and we denote its g.f. by $C(x)$. Consequently, the g.f. $F(x,t)$ can be expressed as follows:

$$F(x,t) = C(x) + t(F(x) - C(x)), \tag{27}$$

since the pattern  can occur at most once in a permutation.

Since the block C may be empty, the g.f. $C(x)$ for the separable permutations avoiding this pattern is given by:

$$C(x) = x(F(x)+1). \tag{28}$$

Substituting (28) into (27), we obtain

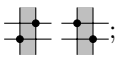
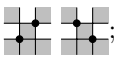
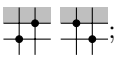
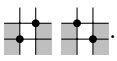
$$F(x,t) = F(x)\big(x+t(1-x)\big) + x(1-t). \tag{29}$$

Furthermore, substituting (1) into (29) yields (26).

To derive the joint distribution of the patterns, we observe that any permutation is either 1, or of the form $1B$ with B nonempty, or of the form $A1$ with A nonempty, or of the form $A1B$ with both A and B nonempty. The corresponding generating functions for these cases are x, $t_1xF(x)$, $t_2xF(x)$, and $t_1t_2\big(F(x)-2xF(x)-x\big)$, respectively, which leads to the desired result. □

4.2 Distributions via functional equations

In this section, we investigate four classes of pairs of mesh patterns whose occurrences in separable permutations may appear more than once, making their distributions considerably more intricate. To determine their distributions, we establish the corresponding functional equations by analyzing all admissible configurations. The pairs are as follows:

S13 =  ; S45 =  ; S61 =  ; S65 =  .

Theorem 17 (Pairs S13 and S14)**.** *Let*

$$F(x,t) = \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{\#(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{\#(\pi)}.$$

Then, $F(x,t)$ satisfies

$$tF^3(x,t) + txF^2(x,t) + ((1+t)x - 1)F(x,t) + x = 0.$$

Proof. Occurrences of the pattern  are precisely the occurrences of ascents, so the result follows from Lemma 1, proved in [8]. □

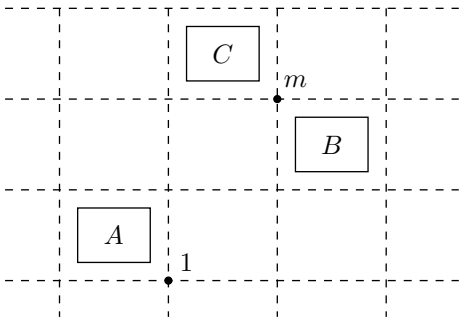


Figure 5: Related to the proof of Theorem 18.

Theorem 18 (Pairs S45–48)**.** *We have*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{\#(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{\#(\pi)} \\ &= \frac{2\left((4(t-1)-1)x^3 + \sqrt{1-6x+x^2} + x\ (f_1(x,t)) + x^2\ (f_2(x))\right)}{(x-3-\sqrt{1-6x+x^2})\,(3+(3-4t)x+\sqrt{1-6x+x^2})}, \end{aligned} \tag{30}$$

where

$$\begin{aligned} f_1(x,t) &= -6 + \sqrt{1-6x+x^2} + 2t\,(1-\sqrt{1-6x+x^2}), \\ f_2(x,t) &= 11 - 4\sqrt{1-6x+x^2} - 2t\,(5-2\sqrt{1-6x+x^2}). \end{aligned}$$

Proof. An occurrence of the pattern  must involve the element 1, and another element m to its right. Moreover, to avoid the pattern 3142, all elements to the left of 1 must be smaller than those to the right of m, as illustrated in Figure 5. Now, the element m is an occurrence of the pattern  in the permutation CmB. Hence, the number of occurrences of the pattern  in the permutation $A1CmB$ is the same as that of the pattern  in CmB. Let $S(x,t)$ denote the g.f. in (6). If $A(x,t)$ is the g.f. of separable permutations with at least one occurrence of  , then

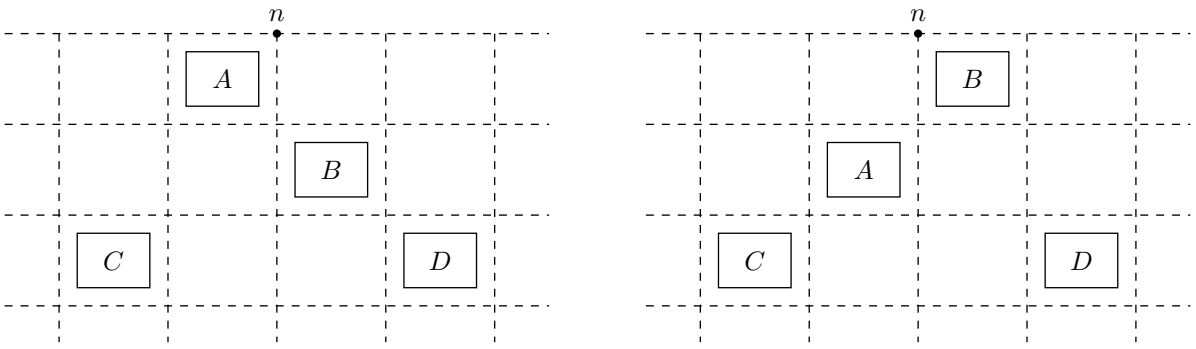
$$A(x,t) = x(S(x,t) - S(x,0))(F(x)+1), \tag{31}$$

where x corresponds to the element 1, and $F(x)+1$ corresponds to the possibly empty block A; note that $S(x,t) - S(x,0)$ is the distribution with at least one occurrence of  .

Adding the contributions from permutations with and without an occurrence, we obtain

$$\begin{aligned} F(x,t) &= x(S(x,t)-S(x,0))(F(x)+1)+\big(F(x)-x(S(x,1)-S(x,0))(F(x)+1)\big) \\ &= F(x)+x(S(x,t)-S(x,1))(F(x)+1)). \end{aligned} \tag{32}$$

Finally, substituting (1) and (6) into (32) yields (30). □



(a) (b)

Figure 6: Related to the proof of Theorem 19.

Theorem 19 (Pairs S61–S64)**.** *We have*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\pi\text{-occ}} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\pi\text{-occ}} \\ &= \frac{x\,\left(x-3+\sqrt{1-6x+x^2}\right)\,\left(tx-3+\sqrt{1-6tx+t^2x^2}\right)}{tx^2-3-\sqrt{1-6x+x^2}-\sqrt{1-6tx+t^2x^2}+f_1(x,t)-x\,f_2(x,t)}, \end{aligned} \tag{33}$$

where

$$\begin{aligned} f_1(x,t) &= \sqrt{1-6x+x^2}\,\sqrt{1-6tx+t^2x^2}, \\ f_2(x,t) &= 1-\sqrt{1-6tx+t^2x^2}+t\,(1-\sqrt{1-6x+x^2}). \end{aligned}$$

Proof. In a permutation $\pi=\pi_1\cdots\pi_n$, the element n, together with each element to its left, forms an occurrence of the pattern  . For $n\geq 2$, separable permutations can be divided into two cases according to the position of the second largest element $n-1$, as illustrated in Figure 6. Specifically, the block A in Figure 6a and the block B in Figure 6b must be nonempty (each containing at least the element $n-1$). We denote by $f_1(x,t)$ and $f_2(x,t)$ the generating functions corresponding to the structures illustrated in Figures 6a and 6b, respectively.

In the situation of Figure 6a, there are three cases to consider.

(a) The blocks B, C, and D are empty, while A is nonempty. In this case, since any element in the block A, together with the element n, contributes an occurrence of the pattern  , the block A can be represented by $F(xt)$, while the element n contributes the term x. Thus, we obtain the term

$$xF(xt).$$

(b) The blocks C and D are empty, while A and B is nonempty. Similar to case (a), the block A can be represented by $F(xt)$, while the element n contributes the term x. Clearly, the elements in the block B are independent of this pattern, and hence B is represented by $F(x)$. Hence, in this case, we have the term

$$xF(xt)F(x).$$

(c) The blocks A and B are nonempty, while among the blocks C and D, at least C is nonempty. In this case, after removing A and B, the element n together with all the remaining elements in π, still is a separable permutation in which the second largest element $n-1$ lies to the left of n; hence, it has the g.f. $f_1(x,t)$. Similar to the previous cases, the block A contributes the term $F(xt)$, while the block B contributes the term $F(x)$. Therefore, this case gives the term

$$F(xt)F(x)f_1(x,t).$$

Combining cases (a)–(c), we have

$$f_1(x,t) = xF(xt) + xF(xt)F(x) + F(xt)F(x)f_1(x,t). \tag{34}$$

When the element $n-1$ is to the right of the largest element n, as shown in Figure 6b, similar to the analysis of $f_1(x,t)$, we obtain

$$f_2(x,t) = xF(x) + xF(xt)F(x) + F(xt)F(x)f_2(x,t). \tag{35}$$

It is clear that $F(x,t)$ can be expressed as follows:

$$F(x,t) = x + f_1(x,t) + f_2(x,t), \tag{36}$$

where the permutation 1 contributes the term x.

Substituting (34) and (35) into (36), we obtain

$$\begin{aligned} F(x,t) &= x + xF(xt) + xF(x) + 2xF(xt)F(x) + F(xt)F(x)\big(f_1(x,t) + f_2(x,t)\big) \\ &= x + xF(xt) + xF(x) + 2xF(xt)F(x) + F(xt)F(x)\big(F(x,t) - x\big) \\ &= x + xF(xt) + xF(x) + xF(xt)F(x) + F(xt)F(x)F(x,t). \end{aligned} \tag{37}$$

Solving this equation for $F(x,t)$ and substituting (1) yields (33). □

Theorem 20 (Pairs S65–S68)**.** *We have*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} \\ &= \frac{3 - x - 4tx + \sqrt{1-6x+x^2} - 4\,f(x,t)}{2t\,\left(1 + x - \sqrt{1-6x+x^2}\right)}, \end{aligned} \tag{38}$$

where

$$f(x,t) = \sqrt{-1 - x + \sqrt{1-6x+x^2} + \tfrac{1}{16}\left(-5 + (4t-1)x + \sqrt{1-6x+x^2}\right)^2}.$$

Proof. Let ab be an occurrence of the pattern  , so that all elements to the right of b and all elements to the left of a are larger than b. Therefore, b is a right-to-left minimum, and a is the leftmost element

smaller than b among those to its left. In other words, in a separable permutation, each right-to-left minimum together with the leftmost element smaller than it, if such element exists, contributes one occurrence of the pattern  . For every right-to-left minimum $b \neq 1$, there exists a unique smaller element a to its left such that ab forms an occurrence of the pattern  . Consequently, in each separable permutation, the number of occurrences of the pattern  corresponds to the number of right-to-left minima minus one.

According to Theorem 1, the distribution of right-to-left minimum on separable permutations is given by the function $G(x,t)$ in (3). Therefore, we have

$$F(x,t) = \sum_{n\geq 1} x^n \sum_{\pi \in S_n(2413,3142)} t^{\mathrm{rmin}(\pi)-1} = \frac{G(x,t)}{t}. \tag{39}$$

By substituting (3) into (39), we obtain (38). □

4.3 Distributions via reducible and irreducible decomposition

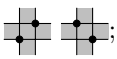
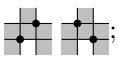
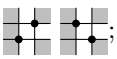
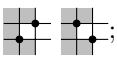
For separable permutations of length $n \geq 2$, each permutation is either reducible or irreducible, as illustrated in Figure 7, where all four blocks A_1, A_2, B_1, and B_2 are nonempty. In the reducible case (see Figure 7a), the block A_1 is irreducible (possibly, of length 1), and A_2 can be any separable permutation. In the irreducible case (see Figure 7b), the block B_1 is either reducible or has a single (largest) element, while B_2 can be any separable permutations. In the proofs of the theorems below, we let $I(x,t)$ and $R(x,t)$ denote, respectively, the generating functions for irreducible and reducible separable permutations, where the variable t records occurrences of the pattern in question.

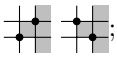
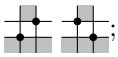
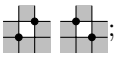
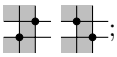


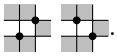
(a) Reducible permutation (b) Irreducible permutation

Figure 7: Structural decomposition of separable permutations of length $n \geq 2$.

In this section, we investigate the distributions of eight classes of pairs of mesh patterns whose distributions are difficult to derive through a direct configuration analysis as in the previous section. Instead, by fully exploiting the structural decomposition of separable permutations, we obtain functional equations that give their distributions. The pairs are as follows:

S5 =  ; S33 =  ; S37 =  ; S41 =  ;

S53 =  ; S57 =  ; S69 =  ; S73 =  ;

S77 =  .

Theorem 21 (Pair S5)**.** *We have*

$$F(x,t) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)}$$
$$= \frac{tx-1+\sqrt{(tx-1)^2-4x\,(1+x-tx)}}{2\,(-1-x+tx)}. \tag{40}$$

Proof. Each ascent $a(a+1)$ formed by consecutive elements contributes exactly one occurrence of the pattern  .

In the irreducible case (see Figure 7b), the pattern  cannot occur across blocks B_1 and B_2. Moreover, any occurrence of the pattern inside B_1 or B_2 is an occurrence of the pattern in the entire permutation. Consequently, $I(x,t)$ can be expressed as

$$I(x,t) = x + (x + R(x,t))\,F(x,t), \tag{41}$$

where x corresponds to the permutation 1, $F(x,t)$ corresponds to the block B_2, and the factor $x+R(x,t)$ represents the g.f. for the block B_1.

In the reducible case (see Figure 7a), the shadings of the pattern  ensure that any occurrence of the pattern within either block A_1 or A_2 is also an occurrence of the pattern in the entire permutation. In addition, if A_1 has a single element and A_2 ends with the largest element, then an extra occurrence of the pattern arises. If $g(x,t)$ denotes the distribution of  on separable permutations ending with the largest element, then in the reducible case we have

$$R(x,t) = (I(x,t)-x)F(x,t) + xtg(x,t) + x(F(x,t)-g(x,t)) \tag{42}$$

where $I(x,t)-x$ records irreducible permutations of length at least 2, and $F(x,t)-g(x,t)$ is the distribution of the pattern on separable permutations not ending with the largest element. Now, we have

$$g(x,t) = x + xtg(x,t) + x(F(x,t)-g(x,t)),$$

where the terms correspond, respectively, to the permutation 1, the separable permutations of length $n\geq 2$ ending with $(n-1)n$, and the separable permutations of length $n\geq 2$ ending with an for $a\neq n-1$. Therefore,

$$g(x,t) = \frac{x\,(F(x,t)+1)}{1+x(1-t)}. \tag{43}$$

Finally, using (41) and (42), we have

$$\begin{aligned} F(x,t) &= I(x,t) + R(x,t) \\ &= F(x,t)\big(I(x,t)+R(x,t)\big) + xF(x,t) + xtg(x,t) - xg(x,t) + x \\ &= F(x,t)\big(F(x,t)+x\big) + x(t-1)g(x,t) + x. \end{aligned} \tag{44}$$

By substituting (43) into (44) and solving for $F(x,t)$, we obtain (40). □

Theorem 22 (Pairs S33–S36)**.** *We have*

$$F(x,t) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)}$$
$$= \frac{2x\,(1+(t-1)\,x)}{1-x-2\,(t-1)\,x^2+\sqrt{1-6x+x^2}}. \tag{45}$$

Proof. In the irreducible case (see Figure 7b), the shading of the lower-right corner of the pattern  prevents this pattern from occurring in B_1. On the other hand, any occurrence of the pattern in B_2 is clearly an occurrence in the entire permutation, and the pattern cannot occur across B_1 and B_2. Therefore, also considering the permutation 1, we have

$$I(x,t) = x + F(x,t)\frac{F(x)+x}{2}. \tag{46}$$

Here, the factor $F(x,t)$ corresponds to the block B_2, while the factor $\frac{F(x)+x}{2}$ represents the generating function for reducible separable permutations corresponding to the block B_1, bearing in mind that B_1 may have length 1.

In the reducible case (see Figure 7a), the shading of the lower-left corner of the pattern  prevents this pattern from occurring in A_1. On the other hand, any occurrence of the pattern in A_2 is clearly an occurrence in the entire permutation. Additional occurrences of the pattern can only arise in the situation when A_2 begins with its largest element a, and A_1 is of length 1 (that is, $A_1 = a+1$ and $a(a+1)$ is an occurrence of ). Since the g.f. for A_2 beginning (resp., not beginning) with the largest element is given by $x(F(x,t)+1)$ (resp., $F(x,t) - x(F(x,t)+1)$), and the g.f. for irreducible separable permutations of length at least 2 is $\frac{F(x)+x}{2} - x$, we have

$$R(x,t) = F(x,t)\frac{F(x)-x}{2} + tx^2(F(x,t)+1) + x(F(x,t) - x(F(x,t)+1)), \tag{47}$$

where the first term corresponds to the case of the irreducible A_1 being of length at least 2. Now using (46) and (47), we obtain

$$\begin{aligned} F(x,t) &= I(x,t) + R(x,t) \\ &= \big(F(x)+x\big)F(x,t) + (t-1)\,x^2\,(F(x,t)+1) + x \end{aligned} \tag{48}$$

By substituting (1) in (48), we obtain (45). □

Theorem 23 (Pairs S37–S40)**.** *We have*

$$F(x,t) := \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)}$$

$$= \frac{1}{8}\Big(5 + x(1-4t) - \sqrt{1-6x+x^2} - \sqrt{2}\,\sqrt{5 + (1-4t+8t^2)\,x^2 + f_1(x,t)}\Big) \tag{49}$$

where

$$f_1(x,t) = 3\sqrt{1-6x+x^2} - x\Big(6 + \sqrt{1-6x+x^2} + 4t\big(5 - \sqrt{1-6x+x^2}\big)\Big).$$

Proof. Any occurrence of  must involve the leftmost element in a given permutation.

In the irreducible case (see Figure 7b), we cannot have any occurrences of the pattern because of the shaded lower-right and upper-left regions (and clearly no occurrences can happen across B_1 and B_2). Therefore, also considering the permutation 1, we have

$$I(x,t) = \frac{F(x)+x}{2}. \tag{50}$$

In the reducible case (see Figure 7a), if ab is an occurrence of the pattern  , then a is the leftmost element in A_2 and b must lie in A_1 (because the upper-right region of the pattern is shaded). Moreover, b must be a right-to-left maximum in A_1, and conversely, any right-to-left maximum in A_1, together with the leftmost element in A_2, forms an occurrence of the pattern. Since all elements of A_1 are greater than those of A_2, the right-to-left maxima in A_1 are exactly the right-to-left maxima of the entire permutation. Consequently, the number of occurrences of the pattern  is equal to the number of right-to-left maxima in the entire permutation. Therefore, the distribution of reducible separable permutations (of length $n \geq 2$) with respect to this pattern coincides with that of the right-to-left maximum statistic. By Theorem 1, this distribution is

$$R(x,t) = \frac{G(x,t)}{1+G(x,t)} - xt, \tag{51}$$

where $G(x,t)$ is given by (3).

Finally, to obtain (49), we substitute (50) and (51) into $F(x,t) = I(x,t) + R(x,t)$, and then substitute (1) and (3). □

Theorem 24 (Pairs S41–S44)**.** *The formula in Theorem 23 holds for the pairs S41–S44.*

Proof. Any occurrence of  must involve the leftmost element in a given permutation.

In the irreducible case (see Figure 7b), it is clear that if ab is an occurrence of  , then both a and b must belong to B_1, so that B_2 cannot contribute any occurrences of the pattern. Therefore, also considering the permutation 1,

$$I(x,t) = x + (R(x,t) + x)\,F(x), \tag{52}$$

where the factor $F(x)$ corresponds to the block B_2.

In the reducible case (see Figure 7a), the leftmost element in the permutation is in block A_2, and hence the pattern cannot be found within A_1. We consider two subcases:

- The block A_2 begins with the smallest element. That is, $A_2 = 1A_2'$, as shown in Figure 9. Then the element 1, together with every left-to-right maximum in the remainder of the permutation, forms an occurrence of  . The g.f. for this subcase is therefore clearly $xG(x,t)$, where $G(x,t)$ is given by Theorem 1. Indeed, the distribution in question coincides with the distribution of left-to-right maxima in $A_2'A_1$, which can be any separable permutation.
- The block A_2 does not start with the smallest element. Then every occurrence of  , if any, must be entirely in A_2, and the number of choices for A_1 are given by $\frac{F(x)+x}{2}$. Therefore, since A_2' can be empty, in this subcase, the g.f. is

$$\big(F(x,t) - x\,(1+G(x,t))\,\big)\frac{F(x)+x}{2}.$$

Consequently, we obtain

$$R(x,t) = xG(x,t) + \big(F(x,t) - x\,(1+G(x,t))\,\big)\frac{F(x)+x}{2}$$

and

$$F(x,t) \quad = \quad I(x,t) + R(x,t)$$

$$= \quad \big(1+F(x)\big)\Big(x + xG(x,t) + \frac{F(x)+x}{2}\big(F(x,t) - x(G(x,t)+1)\big)\Big). \tag{53}$$

Solving (53) for $F(x,t)$ and substituting into the result (1) and (3) yields (49). □

Theorem 25 (Pairs S53–S56)**.** *We have*

$$\begin{aligned} F(x,t) \quad &:= \quad \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{[pattern]}(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{[pattern]}(\pi)} \\ &= \quad \frac{3 - x(1+4t) + \sqrt{1-6x+x^2} - 4\,f(x,t)}{2t\,\left(1+x-\sqrt{1-6x+x^2}\right)}, \end{aligned} \tag{54}$$

where

$$f(x,t) = \sqrt{-1-x+\sqrt{1-6x+x^2} + \tfrac{1}{16}\left(-5+(4t-1)x+\sqrt{1-6x+x^2}\right)^2}.$$

Proof. Any occurrence of [pattern] must involve the leftmost element in a given permutation.

In the irreducible case (see Figure 7b), the rightmost element lies in the block B_2. Moreover, since all elements in B_1 are larger than those in B_2, the first element required to form the pattern [pattern] cannot occur in B_1. In addition, the block B_2 can be any separable permutation associated with this pattern. Therefore, we have

$$I(x,t) = x + F(x,t)\,\frac{F(x)+x}{2}, \tag{55}$$

where x corresponds to the permutation 1, the factor $F(x,t)$ corresponds to the block B_2, while $\frac{F(x)+x}{2}$ represents the g.f. for reducible separable permutations corresponding to B_1, taking into account that B_1 can also be of length 1.

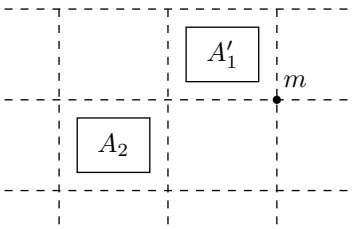


Figure 8: Related to the proof of Theorem 25.

In the reducible case (see Figure 7a), the rightmost element lies in the block A_1. The first element contributing to an occurrence of [pattern] appears in A_2 if and only if the block A_1 ends with its smallest element m, as illustrated in Figure 8. Here, $A_1 = A_1' m$. In this configuration, m cannot form the pattern [pattern] with any element in A_1', but it can form an occurrence of this pattern together with any right-to-left maximum in A_2. Conversely, in this configuration, whenever an occurrence of this pattern appears, its first element must be a right-to-left maximum in A_2. Moreover, A_1' can be any (possibly empty) separable permutation because A_1 ends with m, which guarantees that A_1 is irreducible. Let $A(x,t)$ denote the g.f. associated with this configuration. Then,

$$A(x,t) = x\,G(x,t)\,(F(x)+1)\,, \tag{56}$$

where the block A_2, together with m, contributes the term $xG(x,t)$ ($G(x,t)$ is given by (3)), while $F(x)+1$ corresponds to the block A_1'.

Consequently, we obtain

$$R(x,t) = A(x,t) + ((I(x,t)+x)\ F(x) - A(x,1)), \tag{57}$$

where the factor $F(x)$ corresponds to the block A_2, and the factor $I(x,t)+x$ represents the g.f. for irreducible permutations of length $n \geq 1$ corresponding to the block A_1.

Substituting (55) and (57) into $F(x,t) = I(x,t) + R(x,t)$ yields

$$F(x,t) = x + xF(x) + (F(x)+1)F(x,t)\,\frac{F(x)+x}{2} + x\ (F(x)+1)\,(G(x,t)-F(x))\,. \tag{58}$$

Solving (58) for $F(x,t)$ and substituting (1) and (3) yields (54). □

Theorem 26 (Pairs S57–S60)**.** *The formula in Theorem 25 holds for the pairs S57–S60.*

Proof. Any occurrence of  in a permutation must begin with the smallest element.

In the irreducible case (see Figure 7b), the smallest element lies in B_2, which implies that the pattern  cannot occur within B_1, while B_2 can be any separable permutation. Therefore, we have

$$I(x,t) = x + F(x,t)\,\frac{F(x)+x}{2}, \tag{59}$$

where the factor $F(x,t)$ corresponds to the block B_2, and $\frac{F(x)+x}{2}$ represents the g.f. for reducible separable permutations associated with the block B_1, that can be of length 1.

In the reducible case (see Figure 7a), the smallest element lies in A_2, and A_2 can be any separable permutation. Moreover, the smallest element in A_2, together with any left-to-right maximum in A_1, contributes an additional occurrence of this pattern. Conversely, whenever an occurrence of  appears with its second element in A_1, this element must be a left-to-right maximum in A_1. Consequently, we obtain

$$R(x,t) = F(x,t)\,\frac{G(x,t)}{1+G(x,t)}$$

where $F(x,t)$ represents the contribution of the block A_2, while the term $\frac{G(x,t)}{1+G(x,t)}$ is the distribution of left-to-right maxima on irreducible separable permutations(including the permutation of length $n=1$) given by Theorem 1. Now,

$$F(x,t) = I(x,t) + F(x,t) = x + F(x,t)\,\left(\frac{G(x,t)}{1+G(x,t)} + \frac{F(x)+x}{2}\right). \tag{60}$$

Solving (60) for $F(x,t)$ and substituting (1) and (3) yields (54). □

Theorem 27 (Pairs S69–S72)**.** *We have*

$$\begin{aligned} F(x,t) &:= \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} \\ &= \frac{2x\left(3-x+\sqrt{1-6x+x^2}\right)}{\left(1-x+\sqrt{1-6x+x^2}\right)\left(3+(3-4t)x+\sqrt{1-6x+x^2}\right)}. \end{aligned} \tag{61}$$

Proof. If ab is an occurrence of  then a must be the element 1 in the permutation.

In the irreducible case (see Figure 7b), the bottom-right shading implies that B_1, a reducible separable permutation, contributes no occurrences of  . Since the top-left region is unshaded, every occurrence of  in B_2 is also an occurrence in the entire permutation. As no such pattern can occur across B_1 and B_2, we obtain

$$I(x,t) = x + F(x,t)\frac{F(x)+x}{2}, \tag{62}$$

where $F(x,t)$ corresponds to B_2, the case $B_1 = n$ is considered, and $F(x)$ is given by (1).

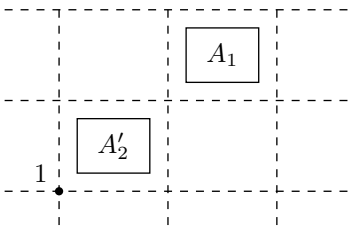


Figure 9: Related to the proof of Theorem 24 and 27.

In the reducible case (see Figure 7a), since the upper-right corner is unshaded and the bottom-left corner is shaded, A_1 cannot contribute any occurrences of  . On the other hand, every occurrence of this pattern in A_2 is also an occurrence in the entire permutation. We then consider two subcases:

- The block A_2 begins with the smallest element. That is, $A_2 = 1A_2'$, as illustrated in Figure 9, where A_2' may be empty. Then the element 1, together with any strict fixed point (i.e., an occurrence of ) in A_1 and A_2', forms an occurrence of  . Since A_1 is an irreducible permutation, such an occurrence of  can appear if and only if A_1 consists of a single point. Therefore, the g.f. for this subcase is

$$tx^2\big(G(x,t)+1\big) \;+\; x\big(1+G(x,t)\big)\frac{F(x)-x}{2},$$

  where $G(x,t)$ is given by Theorem 1. Here the term $tx^2\big(G(x,t)+1\big)$ corresponds to the case where A_1 consists of a single element (i.e., $A_1 = n$), while $x\big(1+G(x,t)\big)\frac{F(x)-x}{2}$ accounts for the case where $|A_1| \geq 2$.

- The block A_2 does not start with the smallest element. Then every occurrence of  , if any, must be entirely in A_2, and the number of choices for A_1 are given by $\frac{F(x)+x}{2}$. Therefore, in this subcase, the g.f. is

$$\big(F(x,t) - x\big(1+G(x,t)\big)\big)\frac{F(x)+x}{2}.$$

Consequently, we obtain

$$R(x,t) = F(x,t)\frac{F(x)+x}{2} + (t-1)x^2\big(1+G(x,t)\big). \tag{63}$$

Substituting (62) and (63) into $F(x,t) = I(x,t) + R(x,t)$, we obtain (61). □

Theorem 28 (Pairs S73–S76)**.** *We have*

$$F(x,t) = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\text{  }(\pi)}$$
$$= \frac{2\left(x+\frac{1}{4}(t-1)\,x\left(1+x-\sqrt{1-6x+x^2}\right)\right)}{1-x+\sqrt{1-6x+x^2}+\frac{1}{2}(1-t)\,x\left(1+x-\sqrt{1-6x+x^2}\right)}. \tag{64}$$

Proof. An occurrence of the pattern  is an ascent satisfying extra conditions.

In the irreducible case (see Figure 7b), since in  the upper-left region is shaded, the block B_2 cannot contain any occurrences of the pattern. Moreover, there are clearly no occurrences spanning both B_1 and B_2. Consequently, the g.f. $I(x,t)$ is

$$I(x,t) = x + \big(R(x,t)+x\big)F(x). \tag{65}$$

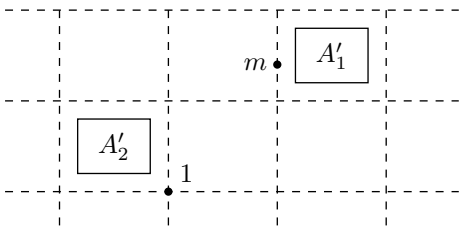


Figure 10: Related to the proof of Theorem 28.

In the reducible case (see Figure 7a), since in  the lower-left region is shaded, the block A_1 cannot contain an occurrence of the pattern, and it does not affect the block A_2. However, an extra occurrence can arise across A_1 and A_2, but only in the situation illustrated in Figure 10, where $A_2 = A_2'1$ and $A_1 = mA_1'$. Let $A(x,t)$ denote the g.f. associated with this configuration. Then

$$A(x,t) = tx\big(F(x,t)+1\big)\frac{F(x)+x}{2}, \tag{66}$$

since $1m$ contributes a factor t, the block A_2 contributes the term $x\big(F(x,t)+1\big)$, and A_1 contributes the term $\frac{F(x)+x}{2}$. Therefore, we have

$$R(x,t) = F(x,t)\,\frac{F(x)+x}{2} + A(x,t) - \frac{A(x,t)}{t}, \tag{67}$$

where the factor $F(x,t)$ corresponds to the block A_2, and the term $\frac{F(x)+x}{2}$ to A_1.

Combining (65)–(67) with $F(x,t) = I(x,t) + R(x,t)$, we obtain

$$F(x,t) = x + xF(x) + (F(x)+1)\left(\frac{F(x)+x}{2}F(x,t) + (t-1)\,x\left(F(x,t)+1\right)\frac{F(x)+x}{2}\right). \tag{68}$$

By solving (68) for $F(x,t)$ and substituting (1), we obtain (64). □

Theorem 29 (Pairs S77–S80)**.** *We have*

$$F(x,t) = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\boxed{\cdot}(\pi)} = \sum_{n\geq 1} x^n \sum_{\pi\in S_n(2413,3142)} t^{\boxed{\cdot}(\pi)}$$
$$= \frac{x\left(3-x-\sqrt{1-6x+x^2}\right)}{1+(t-1)x^2+\sqrt{1-6x+x^2}+f(x,t)} \tag{69}$$

where

$$f(x,t) = x\left(4-\sqrt{1-6x+x^2}-t\left(3-\sqrt{1-6x+x^2}\right)\right).$$

Proof. In the irreducible case (see Figure 7b), because the upper-left region of  is shaded, the block B_2 cannot contain an occurrence of the pattern, and clearly, the pattern does not occur across B_1 and B_2. Therefore, $I(x,t)$ is

$$I(x,t) = x + (R(x,t)+x)\,F(x). \tag{70}$$

In the reducible case (see Figure 7a), since the lower-left region of  is shaded, this pattern cannot occur in A_1, and clearly any occurrence of the pattern in A_2 is also an occurrence in the entire permutation. Moreover, if the irreducible block A_1 has length greater than 1, then there cannot be any occurrences of the pattern spanning both A_1 and A_2. If, on the other hand, A_1 has length 1, then the element 1 in A_2 together with A_1 forms an occurrence of  . These two subcases give, respectively, the first and second summands in

$$R(x,t) = F(x,t)\,\frac{F(x)-x}{2} + txF(x,t). \tag{71}$$

By substituting (70) and (71) into $F(x,t) = I(x,t) + R(x,t)$, we obtain

$$F(x,t) = (1+F(x))\left(F(x,t)\,\frac{F(x)-x}{2} + tx\,F(x,t)\right) + x + xF(x). \tag{72}$$

Solving (72) for $F(x,t)$ and substituting (1), we obtain (69). □

5 Conclusion

In this paper, we first determined the distributions of all mesh patterns of length 1 (see Table 2 and the references therein). In one instance, Theorem 7, we derived the joint equidistribution of strict fixed points and reverse strict fixed points. Using computer experiments and symmetry operations, we then classified the mesh patterns 12 and 21 with identical shadings into 38 equivalence classes based on their distributions; these classes are listed in Tables 3–6. Among the 24 (joint) distributions we derived, several were identical, allowing us to merge some equivalence classes and reduce the total number to 31. These mergings are summarized in Table 7, where patterns with identical distributions are grouped within the same block. Computer experiments confirm that the first several terms of the distributions of the 31 equivalence classes (17 of which we enumerated) are distinct, so no further merging is possible. Enumerating the remaining 14 classes – those missing references in Tables 3–6 – remains an open problem, providing a natural direction for future work.

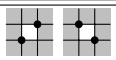
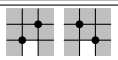
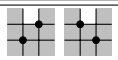
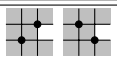
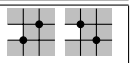
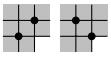
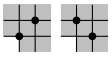
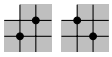
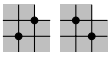
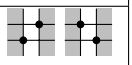
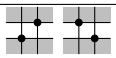
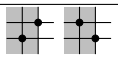
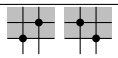
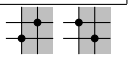
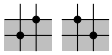
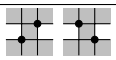
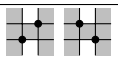
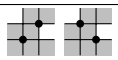
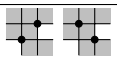
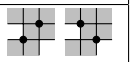
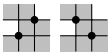
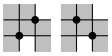
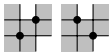
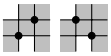
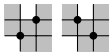
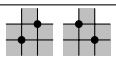
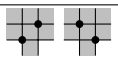
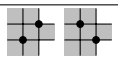
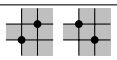
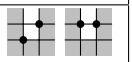
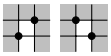
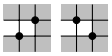
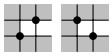
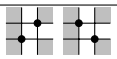
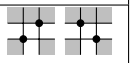
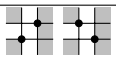
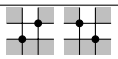
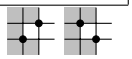
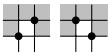
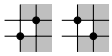
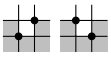
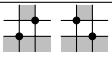
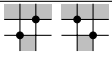
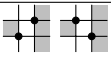
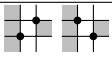
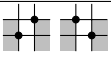
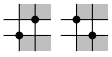
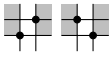
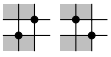
Nr.	Patterns	Nr.	Patterns	Nr.	Patterns	Nr.	Patterns	Nr.	Patterns
S2		S17		S18		S19		S20	
S113		S114		S115		S116		S9	
S10		S29		S30		S31		S32	
S11		S12		S117		S118		S119	
S120		S121		S122		S123		S124	
S21		S22		S23		S24		S25	
S26		S27		S28		S37		S38	
S39		S40		S41		S42		S43	
S44		S57		S58		S59		S60	
S65		S66		S67		S68			

Table 7: Equidistribution classes after further merging.

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